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$$\varepsilon_k^{HF} = \frac{\hbar^2 k^2}{2m} - \frac{2e^2}{\pi} F(k/k_F)$$

$$F(x) = \frac{1}{2} + \frac{(1-x^2)}{4x} \ln\left|\frac{1+x}{1-x}\right|$$

$x = k/k_F$

non-interacting energy at k_F

$$\varepsilon_0^F = \frac{\hbar^2 k_F^2}{2m}$$

$$\varepsilon_k^{HF} / \varepsilon_0 F = x^2 - \frac{4 e^2 m}{\hbar^2 k_F \pi} F(x)$$

$x = k/k_F$ x is dimensionless

$\frac{\hbar^2 k^2}{2m}$ kinetic energy

$\frac{\hbar^2 c^2 k^2}{2 m c^2}$ $\hbar c = 197\ eV\,nm$

$\hbar^2 c^2 \sim eV^2 nm^2$

Bohr radius $a_0 = \frac{\hbar^2}{e^2 m}$

$$n = \frac{N}{\Omega} = 2 \int_0^{k_F} k^2 dk \int d\theta \sin\theta \times \frac{1}{(2\pi)^3} \int d\varphi$$

$$= \frac{k_F^3}{3\pi^2}$$

$$\int d\vec{k} = \int_0^{\infty} dk\, k^2 \int_0^{\pi} d\theta \sin\theta \int_0^{2\pi} d\varphi$$

Total volume : $N \cdot \Omega_e$

$\Omega_e = 4\pi a^3/3$

$$n = \frac{N}{N\Omega_e} = \frac{N}{\Omega} = \frac{3}{4\pi r_s^3} = \frac{k_F^3}{3\pi^2}$$

$$\varepsilon_k^{HF}/\varepsilon_0^{HF} = x^2 - \left(\frac{r_s}{a_0}\right) 0.663\, F(x)$$

$$\langle \Phi_0^{HF} | \mathcal{H} | \Phi_0^{HF} \rangle$$

$$\langle \Phi_0^{HF} | T | \Phi_0^{HF} \rangle =$$

$$\frac{\Omega \hbar^2 k_F^5}{\pi^2 10 m} \qquad \frac{N}{\Omega} = k_F^3/3\pi^2$$

$$\langle T \rangle = \frac{\hbar^2}{10\pi^2 m}\left(\frac{3\pi^2 N}{\Omega}\right)^{5/3}$$

$$= \frac{3\hbar^2}{10m}(4\pi)^{2/3} N/r_s^3$$

$$a_0 = \hbar^2/me^2$$

$$1\,Ry = \frac{me^4}{2\hbar^2} = 13.6\,eV$$

$$\frac{\langle T \rangle}{N} = 2.21\left(\frac{a_0}{r_s}\right) \times 1Ry$$

$$\langle \Phi_0^{HF} | V_{ee} | \Phi_0^{HF} \rangle$$

$$\hat{V}_{ee} = \frac{1}{2} \sum_{\vec{k}_1 \vec{k}_2 \vec{k}_3 \vec{k}_4} \langle \vec{k}_1 \sigma_1 \vec{k}_2 \sigma_2 | \hat{v} | \vec{k}_3 \sigma_3 \vec{k}_4 \sigma_4 \rangle$$

$$\sigma_1 \sigma_2 \sigma_3 \sigma_4$$

$$\times a^{+}_{\vec{k}_1 \sigma_1} a^{+}_{\vec{k}_2 \sigma_2} a_{\vec{k}_4 \sigma_4} a_{\vec{k}_3 \sigma_3}$$

$$\langle \vec{k}_1 \sigma_1, \vec{k}_2 \sigma_2 | v | \vec{k}_3 \sigma_3 \vec{k}_4 \sigma_4 \rangle =$$

$$\lim_{\mu \to 0^+} \frac{e^2}{\Omega} \delta_{\sigma_1 \sigma_3} \delta_{\sigma_2 \sigma_4} \delta_{\vec{k}_1 + \vec{k}_2, \vec{k}_3 + \vec{k}_4}$$

$$y = |\vec{y}| \times \int d\vec{y} \, e^{i(\vec{k}_2 - \vec{k}_4)\vec{y}} \frac{e^{-\mu y}}{y}$$

$$\vec{y} = \vec{x}_1 - \vec{x}_2 \qquad \vec{q} = \vec{k}_2 - \vec{k}_4$$

$$\lim_{\mu \to 0^+} \frac{e^2}{\Omega} \delta_{\vec{k}_1+\vec{k}_2,\, \vec{k}_3+\vec{k}_4} \times \frac{4\pi}{\mu^2+q^2}$$

$$q^2 = |\vec{k}_2 - \vec{k}_4|^2$$

$$\vec{q} = \vec{k}_2 - \vec{k}_4 \quad \wedge \quad \vec{k}_1+\vec{k}_2 = \vec{k}_3+\vec{k}_4$$

$$\vec{k}_1 = \vec{k}_3 - \vec{q} \quad \wedge \quad \vec{k}_3 = \vec{p}$$

$$\vec{k}_4 = \vec{k}$$

$$\hat{V} = \frac{1}{2} \sum_{\sigma_1 \sigma_2} \sum_{\vec{p},\vec{k},\vec{q}} \frac{e^2}{\Omega^2} \frac{4\pi}{\mu^2+q^2} \times a^+_{\vec{k}+\vec{q},\sigma_1} a^+_{\vec{p}-\vec{q}\,\sigma_2} a_{\vec{k}\sigma_2} a_{\vec{p}\sigma_1}$$

$$\Sigma_{\vec{p}\vec{q}\vec{k}} = \Sigma_{\vec{p}\vec{k}\ q=0} + \Sigma_{\vec{p}\vec{k}\ \vec{q}\neq 0}$$

$$(i) \quad \langle \Phi_0^{HF} | V_{ee}(q=0) | \Phi_0^{HF} \rangle$$

$$= \frac{e^2 4\pi}{2\Omega\mu^2}\left(N^2 - N\right)$$

This cancels (lim $N \to \infty$) the term from the ion background + electron–ion interaction

(ii')

$$\langle \Phi_0^{HF} | \sum_{\sigma_1\sigma_2} \sum_{\substack{\vec{p}\vec{k} \\ \vec{q}\neq 0}} \frac{e^2}{2\Omega} \frac{4\pi}{\mu^2+q^2}$$

$$\times a^+_{\vec{p}+\vec{q}\sigma_1} a^+_{\vec{k}-\vec{q}\sigma_2} a_{\vec{k}\sigma_2} a_{\vec{p}\sigma_1} | \Phi_0^{HF} \rangle$$

$$\delta_{\sigma_1\sigma_2} \quad \delta_{\vec{p}+\vec{q},\vec{k}} \quad \delta_{\vec{k}-\vec{q},\vec{p}}$$

$$\vec{k}-\vec{q}=\vec{p} \;\wedge\; \vec{p}+\vec{q}=\vec{k}$$

$$\Rightarrow \quad -2 \cdot \frac{2\pi e^2}{\Omega} \sum_{\substack{\vec{p}\vec{k} \\ \vec{k}\neq\vec{p}}} \frac{1}{|\vec{p}-\vec{k}|^2}$$

taken $\lim_{\mu\to 0}$

$$\frac{1}{\Omega}\sum_{\vec{k}} \rightarrow \frac{1}{(2\pi)^3}\int d^3k$$

$$\int d^3k = \int_0^{k_F} k^2 dk \int_0^{\pi} d\theta \sin\theta \int_0^{2\pi} d\varphi$$

$$-\frac{4\pi e^2 \Omega}{(2\pi)^6}\int d^3k \int d^3p \frac{1}{|\vec{p}-\vec{k}|^2}$$

$$\int d^3k \int d^3p \frac{1}{|\vec{p}-\vec{k}|^2}$$

$$\int d^3p \int d^3k \; \frac{1}{p^2+k^2-2kp\cdot\underbrace{\cos\theta}_{u}}$$

$$s = k/p$$

$$\int d^3p \int d^3k \; \frac{1}{p^2} \; \frac{1}{1+s^2-2su}$$

$p \neq k \qquad p > k \qquad s < 1$

$p < k \qquad s > 1$

$$\frac{1}{\sqrt{1+s^2-2su}} = \sum_l s^l P_l(u)$$

↑ Legendre polynomials

$$\frac{1}{1+s^2-2su} = \sum_{l\lambda} s^{l+\lambda} P_l(u) P_\lambda(u)$$

$$\int_0^{k_F} dk\, k^2 \, 2\pi \int_{-1}^{1} du \times \sum_{l\lambda} \left(\frac{k}{p}\right)^{l+\lambda} \frac{1}{p^2}$$

$(k<p)$

$$\left(u = \cos\theta \quad d\theta \sin\theta = du \right)$$

$$\times P_l(u) P_\lambda(u)$$

$$\left(+ \int_{p<k} d^3p \int_{k\le k_F} \cdots \right)$$

$$\int_{-1}^{1} P_\ell(u)\, P_{\ell'}(u)\, du = \frac{2}{2\ell+1}\,\delta_{\ell\ell'}$$

which gives

$$4\pi \int_{p\le k_F} d^3p \int_{k<p} dk\, k^2 \sum_\ell \left(\frac{k}{p}\right)^{2\ell+2} \times \frac{1}{2\ell+1}$$

with 2nd integral

$$= 8\pi \int_0^{k_F} dp\, p \sum_\ell \frac{1}{(2\ell+3)(2\ell+1)}$$

$$\sum_{l=0}^{\infty} \frac{1}{(2l+1)(2l+3)} =$$

$$\frac{1}{2} \sum_{l=0}^{\infty} \left(\frac{1}{2l+1} - \frac{1}{2l+3} \right)$$

$$= \frac{1}{2} + \frac{1}{2} \sum_{l=0}^{\infty} \left(\frac{1}{2l+3} - \frac{1}{2l+5} \right)$$

$$= 1/2$$

Collecting:

$$\langle \Phi_0^{HF} | V_{ee} | \Phi_0^{HF} \rangle$$

$$= -\frac{e^2 \Omega}{4\pi^3} k_F^4$$

$$\left[n = N/\Omega \;\wedge\; \frac{4\pi r_s^3}{3} = \frac{1}{n} \right]$$

$$\left[r_s = \left(\frac{3}{4\pi n}\right)^{1/3} \;\wedge\; k_F^3 = 3\pi^2 n \right]$$

$$= -\frac{e^2 \Omega}{4\pi^3} \left(3\pi^2 \frac{N}{\Omega}\right)^{4/3} n^{4/3}$$

Divide by N $\quad \frac{1}{n}$

per particle (already v)

$$-\frac{e^2}{4\pi}\,3\,(3\pi^2)^{1/3}\,n^{1/3}$$

$$= -\text{const}\; n^{1/3}$$

Forms the basis for the local density approximation in Density functional theory (DFT)

HF :

$E_{HF}[\Phi_0]$

where $|\Phi_0\rangle$ is a Slater det

variations

$$|\Phi_0\rangle \rightarrow |\Phi_0\rangle + |\delta\Phi_0\rangle$$

$$\left(|\delta\Phi_0\rangle = \left(\sum_{ai} \delta c_i^a a_a^\dagger a_i\right)|\Phi_0\rangle\right.$$

$$\frac{\delta E[\Phi_0]}{\delta \Phi_0} = 0$$

DFT : energy is a functional of the one body density

n

$$E[n] \quad \text{s.t.} \quad \frac{\delta E[n]}{\delta n} = 0$$

$$\hat{H} = \hat{T} + \hat{V}_{ext} + \hat{V}_{int} + \cancel{A_{add}}$$

$$E[n] = \hat{T}[n] + \hat{V}_{ext}[n] + \hat{V}_{int}[n]$$

in 2nd quantization

$$\hat{N} = \sum_p a_p^+ a_p$$

$$\langle \psi | \hat{N} | \psi \rangle = N \text{ (number of particles)}$$

in coordinate space

$$\hat{n}(\vec{r}) = \sum_{i=1}^{N} \delta(\vec{r} - \vec{r}_i)$$

expectation value

$$n(\vec{r}) = \langle \psi | \hat{n}(\vec{r}) | \psi \rangle$$

$$\langle \psi | \psi \rangle = 1$$

$$n(\vec{r}) = \sum_{i=1}^{N} \int d\vec{r}_1 \cdots d\vec{r}_N \times |\psi(\vec{r}_1 \vec{r}_2 \cdots \vec{r}_N)|^2 \delta(\vec{r} - \vec{r}_i)$$

consider the $i = 1$ term and integrate over $\vec{r}_1$

$$n(\vec{r}_1) = \int d\vec{r}_2 \cdots d\vec{r}_N \times |\psi(\vec{r}_1, \vec{r}_2, \cdots \vec{r}_N)|^2$$

This is the probability density function for the coordinate labeled "particle 1"

$$n(\vec{r}) = \sum_{i=1}^{N} \int d\vec{r}_1 \cdots d\vec{r}_N \, \delta(\vec{r} - \vec{r}_i) \times |\psi|^2$$

$$n(\vec{r}) = N \int d\vec{r}_2 \cdots d\vec{r}_N \, |\psi|^2$$

$$\int d^3 r \, n(\vec{r}) = N$$

$$\hat{V}_{ext} = \sum_{i=1}^{N} v_{ext}(\vec{r}_i)$$

$$\langle \psi | \psi \rangle = 1$$

$$\langle \hat{V}_{ext} \rangle = \sum_{i=1}^{N} \int d\vec{r}_1 \, d\vec{r}_2 \cdots d\vec{r}_N$$

$$\times |\psi|^2 v_{ext}(\vec{r}_i) =$$

$$N \int d\vec{r}_1 \cdots \int d\vec{r}_N \, |\psi|^2 v_{ext}(\vec{r}_1)$$

$$= \int d\vec{r}_1 \, v_{ext}(\vec{r}_1) \left[N \int d\vec{r}_2 \, d\vec{r}_3 \cdots d\vec{r}_N |\psi|^2 \right]$$

$$n(\vec{r}_1)$$

$$= \int d\vec{r}\, v_{ext}(\vec{r})\, n(\vec{r})$$

$$= \hat{V}_{ext}[n]$$

$$V_{int}[n] = V_H[n] + E_{xc}[n]$$

$$V_H = \int d\vec{r}_1 \int d\vec{r}_2\, n(\vec{r}_1)\, n(\vec{r}_2)\, v(\vec{r}_1, \vec{r}_2)$$

Local density

$$E_{xc}[n] \sim n^{1/3}$$ (electron spin)